

Study of Stochastic Invertible Multiplier Designs

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Abstract—Invertible logic is a method of reversible computing that can propagate a circuit signal backward from outputs. For example, not only multiplication (forward operation) but also division and factorization (backward operation) can be performed by this circuit style of a single multiplier. Currently, invertible logic has been able to realize only small-scale circuits. In this paper, we design larger-scale circuits than that reported conventionally, using stochastic computing. Furthermore, we also verify the performance of invertible logic using three different multipliers.

keywords—Stochastic computing, Bidirectional computation, Multiplier, Factorization, Hamiltonian.

I. INTRODUCTION

Recently, invertible logic (*IL*) has attracted attention as a method to perform bidirectional computation [1]. In the most basic form, *IL* is composed of probabilistic elements configured as Boltzmann machines [2]. In other words, it is possible to realize invertible multipliers (*IM*)s that can perform not only multiplication (forward operation) but also division and factorization (backwards operations). In conventional works, a magnetic tunnel junction (MTJ) device model has been used as base elements, which realizes only an *IM* with two 2-bit inputs (4-bit Factorizer), based on adder-based multiplier.

In this paper, we design larger *IM*s based on stochastic computing [3], which can be implemented using CMOS gates. In order to investigate the structure of the multiplier optimized for *IM*, three different *IM*s are designed using adder-based, Array [4] and Wallace tree (WT) [5] structure. The performance of *IM*s is evaluated using synthesizable

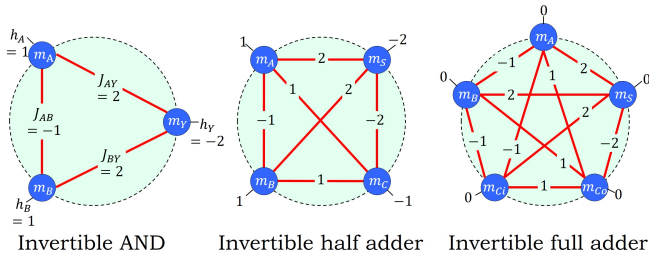


Fig. 1. *IL*s of two-input AND, half adder and full adder. Each node has an individual bias value (h) and interaction weights (J) between the nodes.

SystemVerilog and discussed in terms of computation speed.

II. INVERTIBLE MULTIPLIER

The *IL* is a parallel computational organization of nodes ($m = [-1, +1]$). Fig. 1 shows the *IL*s of two-input AND, half adder and full adder. As shown in this figure, every node is assigned an individual bias value (h) and interaction weights (J) between nodes [2]. Hamiltonian (H) which is the energy function is generally expressed as the following equation by using (h, J):

$$H = -\sum_i h_i m_i - \sum_{i < j} J_{ij} m_i m_j. \quad (1)$$

To design an adder-based *IM* shown in Fig. 2, the h and J of each node can be obtained by superposing (h, J) of computing units. The operation at each node realized by stochastic computing can be expressed by the following two equations:

$$I_i(t+1) = h_i + \sum_j J_{ij} s_j(t) + \text{noise}, \quad (2)$$

$$s_i(t+1) = \text{sgn}\{\tanh(I_i(t+1))\}, \quad (3)$$

where s is $(m+1)/2$ in stochastic unipolar coding. The node-gate circuit is designed as shown in Fig. 3. In this circuit, the multiplication is realized by the left side of multiplexers and the $\tanh$ function is realized by the up/down counter.

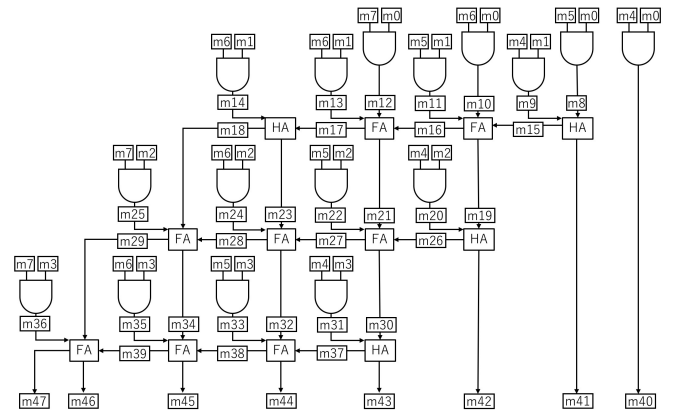


Fig. 2. Structure of a 4-bit adder-based multiplier, where adders are regularly lined up.

TABLE I
COMPARISON OF THREE DIFFERENT INVERTIBLE MULTIPLIERS

Multiplier	Adder-based		Array		WT	
	Forward	Backward	Forward	Backward	Forward	Backward
Logical stages	7		8		5	
Number of nodes	192		192		208	
Correct answer rate [%]	98	97	99	100	99	93
Mean cycle	1728	82024	1565	108486	2243	180087

The adder-based multiplier needs logical stages of $O(n - 1)$ in traditional forward computation, where n is the input bit width of multiplier. In contrast, the logical stages of Array and WT are $O(n)$ and $O(\log_{\frac{3}{2}} n)$, respectively. This makes it possible to increase the speed. On the other hand, the number of nodes used by the WT is larger than that of the other two multipliers due to the structure.

III. SIMULATION RESULTS

We design three *IMs* with two 8-bit inputs using synthesizable SystemVerilog codes. Fig. 4 illustrates how the 4-bit adder-based *IM* performs factorization; converging to correct input values given a fixed output signal. By giving the noise correctly in this way, each node correctly converges while changing the state probabilistically.

Two parameters of “correct answer rate” and “mean cycle” are used for performance evaluation. The correct answer rate indicates the number of corrected patterns out of 100 arbitrary patterns. The mean cycle is the average number of cycles of obtaining the correct outputs.

Table I compares the performance of *IMs* using software and FPGA (Digilent Genesys 2). In terms of the correct answer rate, multiplication can be correctly performed with almost all patterns in both forward and backward operation. In order to obtain the correct outputs, the simulations of forward operation are performed for a maximum of 2^{13} cycles. The simulations of backward operation are performed for a maximum of 2^{20} cycles. Therefore, more cycles are taken to operate backward than forward operation in all *IMs*. Also, WT requires more cycles to obtain correct output

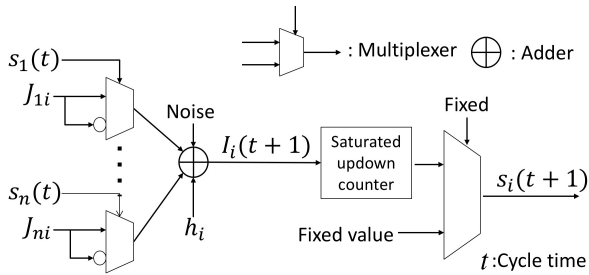


Fig. 3. Node gate circuit based on stochastic computing. The left side of multiplexers realize the multiplication and the up/down counter realizes the $\tanh$ function.

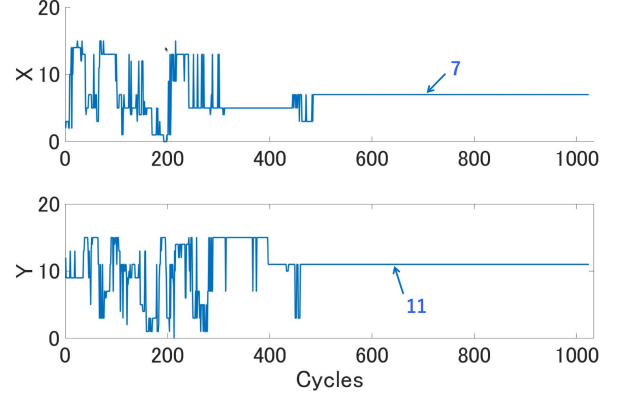


Fig. 4. Factorization of 77 (output) to 7 and 11 (inputs) using 4-bit adder-based *IM*. By gradually decreasing the noise, each node converges to the correct state.

than the other two multipliers. From these results, it would be better to use a multiplier with a small number of nodes, instead of having a small number of logical stages.

IV. CONCLUSION

In this paper, *IMs* have been designed using three multipliers and their performances (both multiplication and factorization) have been compared. As a result, the structure of the multiplier that requires a small number of nodes is suitable for invertible operations. The problem is that the mean cycles of the backward operation are larger than those of the forward one. It is also necessary to correctly recover all patterns. Therefore, we will improve the correct answer rate and reduce the mean cycle by optimizing *IM* structure in the future.

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